A Machine-Checked Itô Calculus for Brownian Motion

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Abstract

We develop the Itô calculus of Brownian motion, machine-checked in Lean 4 over Mathlib and the `BrownianMotion` package. The Itô integral is built on the half-line as a Hilbert-space isometry, from a predictable-rectangle π-system through the density of simple adapted processes; the integral on $[0, T]$ is that map composed with extension by zero. One identity, that the integral at time t is the conditional-expectation projection of its terminal value onto $\mathcal{F}_t$, makes it an L^2 martingale with its time-indexed isometry; a maximal inequality gives it a continuous modification, a local martingale. Its range together with the constants is all of $L^2(\mathcal{F}_T^B)$, which is martingale representation, proved by totality of Doléans exponentials without Malliavin calculus. Itô's formula for $f(t, B_t)$ is proved in L^2 for derivatives of at most exponential growth, with the integrand named; with no growth condition, the compensated process agrees at each time with a continuous local martingale. For Itô processes with continuous adapted coefficients and no bound it holds as an identity of processes, almost surely for all $t \leq T$ at once; the product rule follows. Also proved: Lévy's characterisation, with $M^2 - t$ a local martingale as hypothesis; Girsanov's theorem for bounded predictable drift, concluding a Brownian motion for the filtration when its paths are almost surely continuous; Lipschitz SDEs on every horizon, as an L^2 class; stochastic Fubini. To our knowledge these are the first machine-checked constructions of the Itô integral and proofs of Itô's formula in any proof assistant. The boundary is explicit: the integrator is Brownian motion or an Itô integral against it, and quadratic variation is established in mean, not pathwise. About 34,000 lines of Lean, `sorry`-free, with the axioms of each headline result pinned to Lean's three standard axioms.

1 Introduction

Stochastic calculus sits oddly in the formalization landscape. Its prerequisites (measure-theoretic probability, conditional expectation, L^p spaces, Gaussian measures) are well represented in the major proof-assistant libraries. Its consumers (mathematical finance, stochastic control, stochastic partial differential equations) are fields with strong correctness cultures that cite formal methods as an aspiration. Yet the calculus itself long remained unformalized. Martingales in discrete and continuous time were formalized in Isabelle/HOL by Keskin [10]; the central limit theorem was machine-checked a decade ago [1]; and in 2025 Degenne, Ledvinka, Marion and Pfaffelhuber completed the first formal construction of Brownian motion, in Lean 4 [5, 2]. Their package is the foundation this work builds on, and it is now growing a general theory of stochastic integration: an axiomatic characterisation of the stochastic integral, local martingales, and the Doob–Meyer

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decomposition. The Itô integral of Brownian motion, Itô's formula and the martingale theory of the integral are the layer this paper supplies.

The development is the stochastic-calculus core of a larger library of formalized mathematical finance [4], in which it carries the continuous-time results: Girsanov's theorem, martingale representation and market completeness, stochastic differential equations, and the Black–Scholes pricing that rests on them; the library's no-arbitrage theory is treated separately [3]. Where the library overview surveys breadth, this paper is for readers who want to know exactly what was proved about the calculus, under which hypotheses, and by which proof architecture.

Setting. All results are stated over a probability space $(\Omega, \mathcal{F}, \mu)$ carrying a process $B\colon \mathbb{R}_{\geq 0} \to \Omega \to \mathbb{R}$ whose finite-dimensional laws are those of Brownian motion (the pre-Brownian interface `IsPreBrownianReal`, developed in the `BrownianMotion` package and now part of Mathlib), with measurable coordinates and, where stated, continuous paths. $(\mathcal{F}_t)$ is the natural filtration of B (`natFiltration`).

Contributions.

(1) *The Itô integral as an isometry* (Section 5). The integrand space H_T is the L^2 space of the product measure on $(0, T] \times \Omega$ trimmed to the predictable σ-algebra; simple adapted processes are dense in it, and by exhaustion in the analogous space on the half-line. The integral on the half-line (`itoIntegralL2`) is the unique continuous linear extension of the elementary integral, an isometry, and $\mathcal{I}_T$ (`itoIntegralCLM`) is that map composed with extension by zero.

(2) *The integral as a process* (Section 6). The L^2 process is defined as the conditional projection $(\varphi \bullet B)_t = \mathbb{E}[\mathcal{I}_T(\varphi) \mid \mathcal{F}_t]$ of the terminal integral; adaptedness, the martingale property, the contraction bound, the terminal and time-indexed Itô isometries and L^2-continuity in t are corollaries.

(3) *Itô's formula in L^2* (Section 7). For $f(t, x)$ whose partial derivatives $f_t, f_x, f_{xx}, f_{tt}, f_{tx}, f_{xxx}$ are bounded, and then for those growing at most like $e^{\lambda|x|}$,

$$f(T, B_T) - f(0, B_0) = \mathcal{I}_T\big(f_x(\cdot, B)\big) + \int_0^T \big(f_t + \tfrac{1}{2} f_{xx}\big)(s, B_s)\,\mathrm{d}s \quad \mu\text{-a.s.},$$

with the integrand of $\mathcal{I}_T$ named as the class of $(s, \omega) \mapsto f_x(s, B_s(\omega))$, by a discrete-to-continuous argument through weighted quadratic variation with explicit L^2 remainder bounds and, for the growth case, a cutoff limit carried by the isometry.

(4) *The pathwise integral* (Section 8). The integral process has an almost surely continuous modification whose everywhere-continuous representative is a local martingale for the null-augmented filtration, on $[0, T]$ and on all of $\mathbb{R}_{\geq 0}$.

(5) *Itô's formula without growth conditions* (Section 9). For every f with continuous f, f_t, f_x, f_{xx} and no bound of any kind, the compensated process $f(t, B_t) - f(0, B_0) - \int_0^t (f_t + \frac{1}{2} f_{xx})(s, B_s)\,\mathrm{d}s$ agrees at each t almost surely with a continuous local martingale, by localization at exit times.

(6) *Itô processes* (Section 10). Quadratic variation of an Itô process with adapted volatility, Riemann–Stieltjes sums against an Itô integral, and Itô's formula for $X = x_0 + \int b\,\mathrm{d}s + \int \sigma\,\mathrm{d}B$: with bounded adapted continuous coefficients at a fixed time, with the L^2 integral named; and with continuous adapted coefficients and no bound, as an identity

of processes, almost surely for all $t \leq T$ at once, the stochastic integral being built by gluing clipped integrals. The product rule for two Itô processes follows by polarisation.

(7) *The range and the uniqueness of the integral* (Section 11). Martingale representation: every square-integrable $\mathcal{F}_T^B$-measurable F is $\mathbb{E}[F] + \mathcal{I}_T(\varphi)$ for a unique $\varphi \in H_T$. Integration against $M = \varphi \bullet B$ with its bracket. And agreement with the `BrownianMotion` package's axiomatic stochastic integral.

(8) *What the calculus carries* (Section 12). Lévy's characterisation of Brownian motion; Girsanov's theorem for bounded predictable drifts, concluding that B^θ is a Brownian motion for the natural filtration of B; existence and uniqueness for Lipschitz stochastic differential equations on every horizon; stochastic Fubini for any continuous linear integral, with the Wiener integral recovered as the Itô integral of deterministic integrands; and the Black–Scholes delta hedge as an application.

Section 13 sets the boundary precisely: which parts of the classical theory are present, which are absent, and why the title says "a" calculus rather than "the" calculus.

Changes from the second version. The artifact, pins and sizes are those of release `v1.11.0`. Sections 9–12 are new, as are the growth-condition and process forms of Itô's formula in Section 7. Since the draft of the second version the development has also changed shape. The horizon-T integral is now *defined* from the integral on the half-line, so there is one extension argument rather than one per horizon, and the integral process is defined as the conditional projection of its terminal value, which makes the key identity of Section 6 definitional. The unrestricted formula, which the second version placed beyond the development in its C^2 form, is proved in local-martingale form for f with continuous f_t, f_x, f_{xx}, and for Itô processes the formula is an identity of processes with unbounded continuous adapted coefficients. Lévy's characterisation, the product rule, Girsanov's theorem for the filtration, stochastic differential equations on every horizon and stochastic Fubini are new. Corrections to the second version stand. The bounded-derivative formula names its integrand (the second version displayed an existentially quantified $g_{f'}$ whose prose described more than the statement said); for the same reason the keystone $\int_0^T B \, \mathrm{d}B$ is stated as what the Lean proves, with the named form derived from the growth-condition formula (Section 5.3). The pre-Brownian interface is `IsPreBrownianReal`, now in Mathlib, and the predictable rectangles are consumed from the upstream package. Smaller descriptions are corrected against the source: the σ-algebra the rectangles generate is identified here, not by an upstream lemma; the maximal inequality is Doob's weak-type inequality applied to simple processes; and the closed form of $\int B \, \mathrm{d}B$ preceded the general Riemann bridge rather than consuming it. The scope section is rewritten.

2 Setting and verification discipline

2.1 Foundations consumed

The development is built in Lean 4 [6] (toolchain `v4.35.0`) over Mathlib [11] (commit `e91b128`) and the `BrownianMotion` package [2, 5], whose construction of Brownian motion this work consumes rather than reproves; the artifact pins the package at commit `497e5e2`. The pre-Brownian interface itself lives in Mathlib. From the package the development takes the continuous modification of Brownian motion, the simple processes (`SimpleProcess`) and elementary predictable sets, the predictable rectangles of a filtration and the proof that they form a semiring of sets, the local-martingale predicate `IsLocalMartingale`, Doob's maximal inequality, and the axiomatic stochastic integral of Section 11.3. Time is indexed

by $\mathbb{R}_{\geq 0}$ (Lean's `NNReal`); Lebesgue measure on time enters through `timeMeasure`, restricted to $(0, T]$ by `timeMeasureHorizon`.

2.2 Auditability and reproducibility

Verification follows the library's standing contract [4]. No `sorry` or `admit` occurs anywhere in the development. Two build-enforced audits run `#print axioms` under `#guard_msgs`: a curated one over headline results, and a generated one that pins every library constant a benchmark proof cites. Every numbered theorem of this paper, and every result it names as a headline, is pinned by one of them to `propext`, `Classical.choice` and `Quot.sound`, and a `sorry` or extra axiom beneath any of them fails the build. The `BrownianMotion` package has `sorry`s of its own at the pin (for example in its law of the iterated logarithm and in parts of its stochastic-integration programme); the audit is what certifies that none lies beneath a result reported here. The kernel replay, which re-checks every content module below the elaborator, completes and runs monthly. A hash ledger records the exact source inputs and toolchain pins each benchmark entry was last verified under, so a result cannot go stale silently when a dependency changes. The corpus of the parent library stands at 804 entries, 766 of them `full` and 18 `library_wrapper`; the 20 `reduced_core` entries are not claimed here. The statements displayed below are transcriptions of the Lean statements; where a hypothesis package is abbreviated in prose, the Lean declaration is the authority, and Section A maps every displayed result to its declaration, module and benchmark entry.

2.3 Scale

Counted as the import closure, within the library, of the `Foundations/` modules named in Section A (excluding the attribute module `MathFin/Init.lean`), the Itô tower is 104 modules and about 34,300 lines of Lean, with some 1,030 public theorems; it lies downstream of Mathlib and `BrownianMotion` and upstream of the library's pricing layers. The 26 modules of the second version account for about 8,100 of those lines. Table 1 groups the principal modules by role.

3 The classical theory

We recall the objects the formalization reproduces, in the form they take in the standard references [9, 8], so that the theorems below can be read against them.

Let $(B_t)_{t\geq 0}$ be a Brownian motion on $(\Omega, \mathcal{F}, \mu)$ with natural filtration $(\mathcal{F}_t)$. For a simple adapted process $\varphi_s = \sum_k \xi_k \mathbf{1}_{(t_k, t_{k+1}]}(s)$, with ξ_k bounded and $\mathcal{F}_{t_k}$-measurable, the Itô integral is the finite sum $\int_0^T \varphi \, \mathrm{d}B = \sum_k \xi_k (B_{t_{k+1}} - B_{t_k})$. The defining computation is the *Itô isometry*: because each ξ_k is independent of the forward increment and increments over disjoint intervals are independent,

$$\mathbb{E}\Big[\Big(\sum_k \xi_k (B_{t_{k+1}} - B_{t_k})\Big)^2\Big] = \sum_k \mathbb{E}[\xi_k^2]\,(t_{k+1} - t_k) = \mathbb{E}\int_0^T \varphi_s^2 \, \mathrm{d}s,$$

the cross terms vanishing by adaptedness and the diagonal contributing $\mathbb{E}[\xi_k^2](t_{k+1} - t_k)$ because $\mathbb{E}[(B_{t_{k+1}} - B_{t_k})^2 \mid \mathcal{F}_{t_k}] = t_{k+1} - t_k$. The map $\varphi \mapsto \int_0^T \varphi \, \mathrm{d}B$ is thus an isometry from the simple processes, carrying the $L^2(\lambda \otimes \mu)$ norm on the predictable side, into $L^2(\mu)$; as the simple processes are dense in the predictable L^2 space, it extends to a linear isometry on all of it (Section 5).

As a function of its upper limit, $t \mapsto \int_0^t \varphi \, \mathrm{d}B$ is a continuous L^2 martingale with $\mathbb{E}[(\int_0^t \varphi \, \mathrm{d}B)^2] = \mathbb{E}\int_0^t \varphi_s^2 \, \mathrm{d}s$ (Sections 6 and 8). The quadratic variation of Brownian motion

Role	Principal modules
Discrete engine	`DiscreteIto`, `DiscreteItoPolynomial`, `ItoSquaringIdentity`
Quadratic variation	`QuadraticVariationL2`, `WeightedQuadraticVariation`, `BrownianQuadraticVariation`, `ItoProcessQV`, `AdaptedQuadraticVariation`
Integral (isometry)	`ItoIntegralL2`, `ItoIntegralHorizonDense`, `ItoIntegralL2Dense`, `ItoIntegralCLM`, `ItoIsometryAdapted`, `ItoIntegralBrownian`, `ItoIntegralCovariation`, `ItoIntegralLocality`
Riemann bridges	`ItoRiemannSum`, `ItoIntegralRiemannBridgeAdapted`, `ItoIntegralRiemannBridgeTD`, `AdaptedProcessToLp`, `ItoIntegralPredictableRiemann`
Integral as process	`ItoIntegralProcess`, `ItoIntegralProcessL2Inf`, `ItoIntegralProcessGeneral`, `ItoIntegralProcessMartingale`, `ItoIntegralProcessIsometry`
Itô's formula in L^2	`ItoFormulaRemainder`, `ItoFormulaSquaredL2`, `ItoFormulaCLM`, `ItoFormulaTD`, `ItoFormulaTDRemainder`, `ItoFormulaProcess`, `ItoFormulaLocalized`, `ItoFormulaGBM`
Pathwise regularity	`ItoIntegralProcessContinuousModification`, `ItoIntegralProcessLocalMartingale`, `ItoIntegralProcessLocalMartingaleGeneral`, `ItoIntegralProcessL2Infinite`, `ItoIntegralProcessLocalMartingaleInfinite`
Localization	`ExitTime`, `ItoFormulaCutoff`, `ItoFormulaUnrestrictedLocMart`
Itô processes	`AdaptedStochasticIntegralFreezing`, `AdaptedRiemannStieltjes`, `ItoFormulaItoProcess`, `ItoFormulaAdaptedTD`, `ItoFormulaAdapted`, `DriftProcessModification`, `ItoIntegralLocalised`, `ItoIntegralLocalisedPredictable`, `ItoIntegralLocalisedLocalMartingale`, `ItoFormulaAdaptedUnbounded`, `ItoFormulaAdaptedProcess`, `ItoProductRule`
Representation	`MartingaleRepresentation`, `DoleansStepRepresentation`, `WienerExponentialTotality`, `BrownianCylinderGeneration`
Integration against $\varphi \bullet B$	`ItoIntegralAgainstMartingale`, `LpMulIsometry`, `PointwiseBracket`, `BracketCompensator`, `StochasticIntegralCharacterisation`
Lévy	`LevyCharacterization`, `LevyExpMartingale`, `LevyQuadraticSums`
Girsanov	`ExpMartingaleQBrownian`, `GirsanovConstantTheta`, `GirsanovSimpleTheta`, `GirsanovAdaptedTheta`, `GirsanovPredictableTheta`, `GirsanovFiltered`
SDEs	`ItoProcessPredictable`, `SDEExistence`, `SDEPathwise`, `SDEUniqueness`
Fubini, Wiener	`StochasticFubini`, `StochasticFubiniIto`, `WienerIntegralL2`, `WienerIntegralIto`

TABLE 1. The Itô tower by role: principal modules, all under `MathFin/Foundations/`. The full tower, the import closure of these modules, is just over 100 modules and about 34,300 lines.

is $[B]_t = t$: along partitions of $[0,t]$ with mesh tending to zero, $\sum_k (B_{t_{k+1}} - B_{t_k})^2 \to t$. Itô's formula states that for $f \in C^{1,2}$

$$f(t, B_t) - f(0, B_0) = \int_0^t f_x(s, B_s)\, \mathrm{d}B_s + \int_0^t \left(f_t + \tfrac{1}{2} f_{xx}\right)(s, B_s)\, \mathrm{d}s,$$

the second-order term surviving because $(\mathrm{d}B)^2 = \mathrm{d}t$ in the mean-square sense made exact by $[B]_t = t$. When $f_x(\cdot, B)$ is not square-integrable the stochastic integral is a local martingale, obtained by stopping at the exit times of compact sets (Section 9). Finally, the integral is onto: every centred square-integrable $\mathcal{F}_T^B$-measurable random variable is $\int_0^T \varphi\, \mathrm{d}B$ for a unique predictable φ, the martingale representation theorem (Section 11).

4 Design of the formalization

Four representation choices shape the development and are worth stating before the theorems that rest on them.

The integrand, after the upstream package. The simple integrands are not re-encoded here. `BrownianMotion` provides `SimpleProcess`, a bounded value at time zero together with finitely many bounded values on intervals, each measurable at the interval's left endpoint, and the predictable rectangles of a filtration together with the proof that they form a semiring of sets. This work consumes both: the elementary Itô integral is defined on `SimpleProcess`, the rectangle family is the upstream one, and the π-system property needed for Dynkin's theorem is projected from the upstream semiring. What is built here is the layer above: the identification of the σ-algebra the rectangles generate, the orthogonal-complement density and the isometric extension.

Time as `NNReal`. The time axis is $\mathbb{R}_{\geq 0}$, not $\mathbb{R}$ with a positivity side condition. Increments $B_t - B_s$ then carry $s \leq t$ in their type-level data rather than as a hypothesis to discharge, and a bounded horizon enters through a single finite measure (`timeMeasureHorizon`, of total mass T) rather than through interval hypotheses threaded across lemmas.

A trimmed measure. The integrand space is the L^2 space of a measure that has already been restricted to the predictable σ-algebra, rather than an ambient L^2 space with a sub-σ-algebra measurability side condition. Predictability is then a property of membership in H_T, not a hypothesis threaded through every limit: a Cauchy sequence in H_T has its limit in H_T, so when Itô's formula realizes $s \mapsto f'(B_s)$ as an L^2 limit of step processes the limit's predictability is free. The cost is a one-time investment in the trimmed-measure L^2 API; the saving recurs at every density and convergence argument downstream.

One construction pattern, and one integral. The integral, the integral as a process and the integral against an Itô integral are built by the same idiom: a norm-preserving linear map on a dense subspace, then its unique continuous linear extension (Mathlib's `LinearMap.extendOfNorm`). Every identity proved on simple processes then transfers to all of H_T by continuity and density, through the equalizer property of dense ranges (`DenseRange.equalizer`). The extension is carried out once, on the half-line; the integral on a horizon T is defined as that integral composed with extension by zero (`bandExtend`), and the process on $[0, T]$ likewise. That mechanism is what turns Section 6 into a sequence of short arguments.

5 The L^2 Itô integral

5.1 The integrand space

Fix $T \in \mathbb{R}_{\geq 0}$ and write λ_T for Lebesgue measure on time restricted to $(0, T]$. The predictable rectangles

$$\mathcal{R} = \{(s,t] \times A : \ s < t, \ A \in \mathcal{F}_s\} \cup \{\{0\} \times A_0 : \ A_0 \in \mathcal{F}_0\}$$

of the natural filtration form a π-system (`isPiSystem_predictableRect`) that generates the predictable σ-algebra $\mathcal{P}$ (`generateFrom_predictableRect`, proved here directly); Listing 1 shows the definition, which is the upstream one. The integrand space is

$$H_T = L^2\big((0,T] \times \Omega, \ \mathcal{P}, \ (\lambda_T \otimes \mu) \restriction_{\mathcal{P}}\big),$$

realized as the L^2 space of the product measure trimmed to $\mathcal{P}$ (`trimMeasure`). The analogous space on the half-line, H_∞, uses Lebesgue measure on all of $\mathbb{R}_{\geq 0}$ in place of λ_T (it is

`BrownianMotion`'s `L2Predictable`). The integral on the half-line is a continuous linear map out of H_∞, and $\mathcal{I}_T$ is its composite with extension by zero from H_T (Listing 2).

Listing 1. The predictable rectangles, consumed from `BrownianMotion` *(`Foundations/ItoIntegralHorizonDense`).*

```
def predictableRect (hBmeas : ∀ t, Measurable (B t)) :
    Set (Set (ℝ≥0 × Ω)) :=
  (ItoIntegralL2.natFiltration (mΩ := mΩ) hBmeas).predictableRectangles

lemma isPiSystem_predictableRect (hBmeas : ∀ t, Measurable (B t)) :
    IsPiSystem (predictableRect (mΩ := mΩ) hBmeas) :=
  (MeasureTheory.Filtration.isSetSemiring_predictableRectangles _).isPiSystem
```

Listing 2. The Itô integral on the half-line and on a horizon, and its isometry (`Foundations/ItoIntegralL2Dense`, `ItoIntegralCLM`).

```
noncomputable def itoIntegralL2 (hBmeas : ∀ t, Measurable (B t)) :
    Lp ℝ 2 ((timeMeasure.prod μ).trim
      (natFiltration (mΩ := mΩ) hBmeas).predictable_le_prod) →L[ℝ] Lp ℝ 2 μ :=
  (itoAssembly (μ := μ) hB hBmeas).extendOfNorm (simpleAssembly (μ := μ) hBmeas)

noncomputable def itoIntegralCLM (hB : IsPreBrownianReal B μ) (T : ℝ≥0)
    (hBmeas : ∀ t, Measurable (B t)) :
    Lp ℝ 2 (trimMeasure (μ := μ) T hBmeas) →L[ℝ] Lp ℝ 2 μ :=
  (itoIntegralL2 hB hBmeas).comp (bandExtend T hBmeas).toContinuousLinearMap

theorem itoIntegralCLM_norm (hB : IsPreBrownianReal B μ) (T : ℝ≥0)
    (hBmeas : ∀ t, Measurable (B t))
    (f : Lp ℝ 2 (trimMeasure (μ := μ) T hBmeas)) :
    ‖itoIntegralCLM hB T hBmeas f‖ = ‖f‖
```

5.2 Simple processes, density, isometry

A simple adapted process is a finite combination of indicator rectangles $\xi_s \cdot \mathbf{1}_{(s,t]}$ with ξ_s bounded and $\mathcal{F}_s$-measurable; `TBoundedSP` is the submodule of those supported in $[0, T]$ and `simpleProcessL2Horizon` sends it into H_T. The elementary integral maps such a process to $\sum \xi_s (B_t - B_s)$ (`itoAssemblyHorizon`); independence and the Gaussian increment moments give the elementary Itô isometry (`ItoIntegralL2.assembly_isometry`).

Density of the T-bounded simple processes in H_T (`simpleAssemblyHorizon_denseRange`) is the technical core of the construction and the part of it not inherited from the upstream package. It is proved by the orthogonal-complement method. Take $\psi \in H_T$ orthogonal to every elementary integrand. Testing against the indicator of a rectangle $(a, b] \times F$ shows that the integral of ψ over that rectangle vanishes; the measurable sets on which $\int \psi = 0$ form a λ-system containing the rectangle π-system, so Dynkin's theorem extends the vanishing to all of $\mathcal{P}$ (`setIntegral_eq_zero_of_orthogonal_pred`), forcing $\psi = 0$ almost everywhere. Density in H_∞ follows by exhaustion of the σ-finite time axis by the horizons $T = n$, which is what lets `LinearMap.extendOfNorm` produce the integral `itoIntegralL2` on the half-line, with its isometry `itoIntegralL2_norm`. The horizon-T integral $\mathcal{I}_T$ (`itoIntegralCLM`) is that integral composed with `bandExtend`, so its linearity, continuity and isometry (`itoIntegralCLM_norm`) are those of the two factors, and on a T-bounded simple process it is the elementary integral (`itoIntegralCLM_simpleAssemblyHorizon`). The horizon density is kept because the consumers use it to reduce statements about $\mathcal{I}_T$ to simple

processes.

5.3 The keystone example

The integral's first consumer was the closed form of $\int_0^T B\,\mathrm{d}B$: there is an integrand $g_B \in H_T$, built as the H_T-limit of truncated left-endpoint step processes of $s \mapsto B_s$, with

$$\mathcal{I}_T(g_B) = \tfrac{1}{2}\big(B_T^2 - B_0^2 - T\big) \qquad (\texttt{itoIntegralCLM_brownian}).$$

The Itô correction $-T/2$ emerges as follows: the summation-by-parts identity `discrete_squaring_identity` writes

$$\sum_k B_{s_k}(B_{s_{k+1}} - B_{s_k}) = \tfrac{1}{2}\Big(B_T^2 - B_0^2 - \sum_k (B_{s_{k+1}} - B_{s_k})^2\Big),$$

whose left side is the discretized $\int_0^T B\,\mathrm{d}B$ and whose squared-increment sum converges in L^2 to T (Section 7.2); what is left is $\frac{1}{2}(B_T^2 - B_0^2 - T)$. Because the left-endpoint coefficients B_{s_k} are unbounded Gaussians, the sums are not elementary integrals of simple processes; the proof truncates them to $[-M, M]$ and removes the truncation in the L^2 limit, which is how the Itô integral is defined for unbounded integrands in any case. The general Riemann bridge, which identifies $\mathcal{I}_T$ of a step-process realization of a bounded continuous function of B with the classical partial sums and feeds Itô's formula, came afterwards and builds on this file.

The statement records the integral, not the integrand. The construction makes g_B the class of B, but the theorem does not say so, and by martingale representation (Theorem 11.1) the bare existence of *some* integrand would hold for any centred square-integrable $\mathcal{F}_T^B$-measurable target. The named form, $\mathcal{I}_T(B) = \frac{1}{2}(B_T^2 - B_0^2 - T)$ with B the class of $(s, \omega) \mapsto B_s(\omega)$, is the instance $f(t, x) = x^2$ of the growth-condition formula (Theorem 7.3), whose statement names its integrand: it gives g equal to $2B$ almost everywhere with $B_T^2 - B_0^2 = \mathcal{I}_T(g) + T$. That instance is a few lines from Theorem 7.3 but is not stated separately in the library. The identity remains the smallest statement in which the calculus is visibly non-classical, and it anchors the remainder analysis of Section 7.

5.4 The Wiener integral

The library keeps a Wiener integral on $L^2((0, T])$, built without a filtration for the Gaussian-process and distributional layers. It is the special case of $\mathcal{I}_T$: on deterministic integrands the two continuous linear maps agree, $\texttt{wienerIntegralLp} = \mathcal{I}_T \circ \texttt{detEmbed}$ (`wienerIntegralLp_eq_itoIntegralCLM`), because both are continuous, both send $\mathbf{1}_{(s,t]}$ to $B_t - B_s$, and step indicators are dense. The Wiener isometry therefore follows from `itoIntegralCLM_norm`, the embedding of deterministic integrands being an isometry (`detEmbed`).

6 The integral as a process

6.1 The simple layer

For a simple process V and $t \le T$, the partially summed elementary integral $(V \bullet B)_t$ is adapted (`itoSimpleProcess_adaptedAt`) and a martingale (`itoSimpleProcess_isMartingale`). The martingale step is the conditional vanishing of adapted-weighted future increments: by the characterizing property of conditional expectation it suffices that $\int_A \xi(B_{t_1} - B_{t_0})\,\mathrm{d}\mu = 0$ for every $A \in \mathcal{F}_{t_0}$, and $\xi \mathbf{1}_A$ is again bounded and $\mathcal{F}_{t_0}$-measurable, so the increment, independent of $\mathcal{F}_{t_0}$ with mean zero, factors the integral as $\mathbb{E}[\xi \mathbf{1}_A]\,\mathbb{E}[B_{t_1} - B_{t_0}] = 0$. The martingale proof runs this argument set by set; the conditional

statement it amounts to is recorded on its own in Listing 3. The simple layer also carries the per-time isometry $\mathbb{E}[(V \bullet B)_t^2] = \int_0^t \mathbb{E}[V_s^2]\,ds$ (`itoSimpleProcess_isometry_time`) and L^2-continuity in t (`itoSimpleProcessLp_l2_continuous`); the next subsection lifts all three to general integrands at once.

Listing 3. Adapted weights kill future increments (`Foundations/ItoIntegralProcessMartingale`).

```
theorem condExp_adapted_mul_increment (hBmeas : ∀ t, Measurable (B t))
    {t₀ t₁ : ℝ≥0} (ht : t₀ ≤ t₁) {φ : Ω → ℝ}
    (hφ : Measurable[ItoIntegralL2.natFiltration hBmeas t₀] φ) {C : ℝ} (hC : ∀ ω,
    ↪ |φ ω| ≤ C) :
    μ[fun ω ↦ φ ω * (B t₁ ω - B t₀ ω) | ItoIntegralL2.natFiltration hBmeas t₀]
      =ᵐ[μ] 0
```

6.2 The key identity

For general $\varphi \in H_T$ the process is a continuous linear map (`itoProcessCLM`), and the development's central structural fact is that it is the conditional projection of the terminal integral.

Theorem 6.1 (`itoProcessCLM_eq_condExpL2`). *For every $t \geq 0$ and $\varphi \in H_T$, as elements of $L^2(\Omega, \mu)$,*

$$(\varphi \bullet B)_t = \mathbb{E}\big[\mathcal{I}_T(\varphi) \;\big|\; \mathcal{F}_t\big],$$

the right side realized by Mathlib's L^2 conditional-expectation projection `condExpL2`.

Listing 4. The integral process is the conditional projection of its terminal value (`Foundations/ItoIntegralProcessGeneral`).

```
theorem itoProcessCLM_eq_condExpL2 (T t : ℝ≥0) (hBmeas : ∀ u, Measurable (B u))
    (φ : Lp ℝ 2 (trimMeasure (μ := μ) T hBmeas)) :
    itoProcessCLM hB T t hBmeas φ
      = (condExpL2 ℝ ℝ ((natFiltration hBmeas).le t)
          (itoIntegralCLM hB T hBmeas φ) : Lp ℝ 2 μ)
```

In the present development this is a definition: `itoProcessCLM` is the infinite-horizon process `itoProcessL2Inf`, the $\mathcal{F}_t$-projection of `itoIntegralL2`, composed with `bandExtend`, and the Lean proof of the theorem is `rfl`. The mathematical content is therefore carried by the bridge to the elementary integral: on a T-bounded simple process V, the projection of the terminal integral is the partially summed elementary integral, $\mathbb{E}[\mathcal{I}_T(V) \mid \mathcal{F}_t] = (V \bullet B)_t$ (`itoProcessCLM_simpleAssemblyHorizon`), which is the martingale property of the simple layer. Everything else about the process is a corollary of the definition and standard properties of conditional expectation, proved once for the process on the half-line:

- *adaptedness*, since a `condExpL2` projection lands in the $\mathcal{F}_t$-measurable subspace;
- the *martingale property* (`itoIntegralProcessGen_isMartingale`), by the tower property;
- the *contraction bound* $\|(\varphi \bullet B)_t\|_{L^2} \leq \|\varphi\|_{H_T}$ (`itoProcessCLM_norm_le`), since conditional expectation is a contraction;
- the *terminal isometry* at $t = T$ (`itoProcessCLM_norm_terminal`);
- *L^2-continuity* of $t \mapsto (\varphi \bullet B)_t$ (`itoIntegralProcessGen_l2_continuous`), by a t-uniform contraction estimate against an approximating simple sequence.

Defining the process as a projection of its terminal value is what turns the martingale theory of the Itô integral into a sequence of corollaries rather than a sequence

of constructions. Two further structural facts follow the same pattern. The isometry polarizes: $\mathbb{E}[\mathcal{I}_T(\varphi)\,\mathcal{I}_T(\psi)] = \int \varphi\psi \,\mathrm{d}(\lambda_T \otimes \mu)$, the covariation of two Itô integrals (`ItoIntegralCovariation`). And the integral is local in time: if φ vanishes on $[0, a]$ and Z is $\mathcal{F}_a$-measurable, then $\mathcal{I}_T(Z\varphi) = Z\,\mathcal{I}_T(\varphi)$ (`ItoIntegralLocality`), the "freeze the past, integrate the future" step behind the Doléans exponentials of Section 11.1.

6.3 The time-indexed Itô isometry

The terminal isometry is the classical $\|\mathcal{I}_T(\varphi)\|_{L^2}^2 = \|\varphi\|_{H_T}^2$. Its refinement to intermediate times, the energy law of the integral as a process, is the second fact every textbook calls "the Itô isometry".

Theorem 6.2 (`itoProcessCLM_norm_sq`). *For every $t \le T$ and $\varphi \in H_T$,*

$$\mathbb{E}\big[(\varphi \bullet B)_t^2\big] = \int_{(0,t]\times\Omega} \varphi^2 \,\mathrm{d}(\lambda_T \otimes \mu) \restriction_{\mathcal{P}} .$$

Listing 5. The time-indexed Itô isometry (`Foundations/ItoIntegralProcessIsometry`).

```
theorem itoProcessCLM_norm_sq {t T : ℝ≥0} (htT : t ≤ T) (hBmeas : ∀ u, Measurable
    ↪ (B u))
    (φ : Lp ℝ 2 (trimMeasure (μ := μ) T hBmeas)) :
    ‖itoProcessCLM hB T t hBmeas φ‖ ^ 2
      = ∫ z in (Set.Ioc 0 t ×ˢ (Set.univ : Set Ω)), (φ z) ^ 2
          ∂(trimMeasure (μ := μ) T hBmeas)
```

The proof is again the equalizer pattern, against a different pair of continuous functionals: $\varphi \mapsto \|(\varphi \bullet B)_t\|^2$ and $\varphi \mapsto \int_{(0,t]} \varphi^2$ agree on simple processes by a band reconciliation that splits the trimmed measure across $(0, t] \cup (t, T]$ (`itoSimpleProcessLp_band_isometry`, `integral_rectTerm_mul_band`). Stating it for general integrands at every t, not only at $t = T$ and not only for simple processes, is what makes the integral as a process a complete object rather than a martingale with a missing variance law.

7 Itô's formula in L^2

7.1 Architecture

The route to Itô's formula is discrete-to-continuous (Figure 1).

1. *Discrete identity.* For any partition, second-order Taylor expansion of f along the increments gives an exact discrete Itô identity (`DiscreteIto`) with an explicit Lagrange remainder.
2. *Weighted quadratic variation.* The term $\sum_k f''(B_{t_k})(\Delta_k B)^2$ converges in L^2 to $\int_0^T f''(B_s)\,\mathrm{d}s$; this is proved for adapted weight processes (`WeightedQuadraticVariation`), reusing the Gaussian moment estimates of the unweighted limit.
3. *Remainder.* The third-order remainder is $O(1/n)$ in L^2 under a bound on f''' (`ItoFormulaRemainder`); the time-dependent variant needs a two-dimensional remainder with mixed partials, also $O(1/n)$ (`ItoFormulaTDRemainder`).
4. *Riemann bridge.* The discretized stochastic term is $\mathcal{I}_T$ applied to a step-process realization of $f'(B)$; every such left-endpoint sum is the one sum `riemann`σ at a particular integrand.
5. *Limit.* The step integrands converge in H_T to the class of $s \mapsto f'(B_s)$. Predictability of the limit is free (the limit is taken inside H_T); its existence for bounded continuous

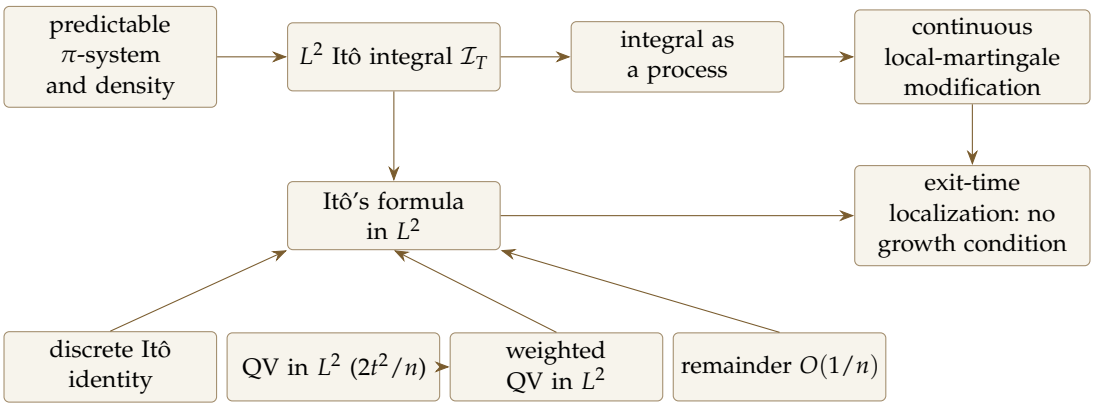


FIGURE 1. Proof architecture. The isometry layer (top) builds the integral, the process and its continuous modification; the discrete-to-continuous layer (bottom) supplies the vanishing terms, which meet at Itô's formula; exit-time localization, through the process form of the formula for Itô processes, removes every growth condition.

data is `itoIntegralCLM_of_bdd_cont_td`, a dominated-convergence argument on the trimmed measure.

7.2 The fluctuation engine: quadratic variation in L^2

The second-order term of Itô's formula is deterministic in the limit for one reason: the squared increments of Brownian motion have vanishing fluctuation in L^2. Along the uniform partition of $[0,t]$ into n pieces of width $\Delta = t/n$,

$$\Big\| \sum_k (B_{s_{k+1}} - B_{s_k})^2 - t \Big\|^2_{L^2(\mu)} = \frac{2t^2}{n} \longrightarrow 0 \qquad (\texttt{sum_increment_sq_sub_sq_integral}, \texttt{tendsto_qv}).$$

The value is exact, not asymptotic: for any partition $0 = s_0 \le \cdots \le s_n$ the mean-square error is $\sum_k 2(s_{k+1} - s_k)^2$ (`sum_increment_sq_sub_sq_integral`), which is $2t^2/n$ for the uniform one. Writing $Y_k = (B_{s_{k+1}} - B_{s_k})^2 - \Delta$ for the centred squared increment, the Y_k over disjoint intervals are independent, so the cross terms vanish and $\mathbb{E}[(\sum_k Y_k)^2] = \sum_k \mathbb{E}[Y_k^2]$. Each term is the Gaussian kurtosis, $\mathbb{E}[Y_k^2] = \mathbb{E}[(B_{s_{k+1}} - B_{s_k})^4] - \Delta^2 = 3\Delta^2 - \Delta^2 = 2\Delta^2$, using $\mathbb{E}[X^4] = 3(\operatorname{Var} X)^2$ for a centred Gaussian (`integral_pow4_gaussianReal`, transported to the increment law by `integral_increment_pow4`). Summing, $n \cdot 2\Delta^2 = 2t^2/n$. The fourth-moment constant 3, and nothing else, is the reason the quadratic variation is t rather than 0; this is the formal content of $(\mathrm{d}B)^2 = \mathrm{d}t$.

7.3 The weighted second-order term

Itô's formula needs this limit carrying a weight: for a bounded adapted process w with continuous paths,

$$\sum_k w_{t_k} (B_{t_{k+1}} - B_{t_k})^2 \longrightarrow \int_0^T w_s \, \mathrm{d}s \quad \text{in } L^2 \qquad (\texttt{tendsto_weighted_qv_process}),$$

instantiated at $w_s = f''(B_s)$ for the autonomous formula and $w_s = f_{xx}(s, B_s)$ for the time-dependent one. The proof splits the difference into a fluctuation term and a Riemann term. The fluctuation term $\sum_k w_{t_k}((B_{t_{k+1}} - B_{t_k})^2 - \Delta_k)$ vanishes in L^2 by the orthogonality-plus-kurtosis estimate above, now carrying the $\mathcal{F}_{t_k}$-measurable bounded weight: adaptedness of w_{t_k} is exactly what keeps the cross terms zero (`integral_adapted_mul_centered_sq`). The

Riemann term vanishes pathwise by continuity of $s \mapsto w_s(\omega)$, then in L^2 by dominated convergence against the bound on w. Neither half uses where the weight came from, only that it is adapted, bounded and path-continuous, which is why one lemma serves the autonomous, the time-dependent, and later the Itô-process formula.

7.4 The remainder

What survives after the first- and second-order terms is a sum of cubic Taylor remainders, and it must vanish in L^2. The per-step remainder R_k obeys $|R_k| \le C_3 |B_{s_{k+1}} - B_{s_k}|^3$ under $|f'''| \le C_3$ (`abs_discreteTaylorRemainder_le`). Hence $\mathbb{E}[R_k^2] = O(\mathbb{E}[(\Delta_k B)^6]) = O(\Delta^3)$ by the Gaussian sixth moment (`integral_pow6_gaussianReal`), and Cauchy–Schwarz across the n terms gives

$$\mathbb{E}\Big[\Big(\sum_k R_k\Big)^2\Big] \le n \sum_k \mathbb{E}[R_k^2] = O\big(n \cdot n \cdot (T/n)^3\big) = O(1/n) \longrightarrow 0.$$

The third order is the only place a third derivative is used. The time-dependent variant runs the same estimate on a two-dimensional Taylor remainder with mixed partials.

7.5 The bounded-derivative formula

Theorem 7.1 (`ito_formula_L2_bddDeriv`). *Let B have measurable coordinates and continuous paths, and let $f : \mathbb{R} \to \mathbb{R}$ be three times differentiable with $|f'| \le C_1$, $|f''| \le C_2$, $|f'''| \le C_3$. Then there is $g \in H_T$ with $g = f'(B_s(\omega))$ for $(\lambda_T \otimes \mu)\restriction_{\mathcal{P}}$-almost every (s, ω), such that μ-almost surely*

$$f(B_T) - f(B_0) = \mathcal{I}_T(g) + \tfrac{1}{2} \int_{(0,T]} f''(B_s)\, \mathrm{d}s.$$

Listing 6. Itô's formula, bounded-derivative class (`Foundations/ItoFormulaCLM`).

```
theorem ito_formula_L2_bddDeriv
    (hBmeas : ∀ t, Measurable (B t)) (hBcont : ∀ ω, Continuous (fun s : ℝ≥0 ↦ B s
    ↪ ω))
    (T : ℝ≥0) {f f' f'' f''' : ℝ → ℝ}
    (hf : ∀ x, HasDerivAt f (f' x) x) (hf' : ∀ x, HasDerivAt f' (f'' x) x)
    (hf'' : ∀ x, HasDerivAt f'' (f''' x) x)
    {C1 : ℝ} (hf1 : ∀ x, |f' x| ≤ C1) {C2 : ℝ} (hf2 : ∀ x, |f'' x| ≤ C2)
    {C3 : ℝ} (hf3 : ∀ x, |f''' x| ≤ C3) :
    ∃ gf' : Lp ℝ 2 (trimMeasure (μ := μ) T hBmeas),
      (⇑gf' =ᵐ[trimMeasure (μ := μ) T hBmeas] fun z ↦ f' (B z.1 z.2)) ∧
      (fun ω ↦ f (B T ω) - f (B 0 ω)) =ᵐ[μ]
        (fun ω ↦ (itoIntegralCLM hB T hBmeas gf') ω
          + (1 / 2) * ∫ s in Set.Ioc 0 T, f'' (B s ω) ∂ItoIntegralL2.timeMeasure)
```

The first conjunct is what makes the second Itô's *formula* rather than a decomposition into some Itô integral: the stochastic term is the integral of the class of $f'(B)$, not of an unnamed element of H_T.

Proof sketch. In Lean the theorem is the case $f_t = 0$ of the time-dependent formula (Theorem 7.2), with no separate proof; the sketch is that proof with the time variable frozen. Fix the uniform partition of $[0, T]$ into n pieces and expand f to second order along each increment. The telescoping sum gives the exact discrete identity $f(B_T) - f(B_0) = S_n + \frac{1}{2} Q_n + R_n$, with $S_n = \sum_k f'(B_{s_k})(B_{s_{k+1}} - B_{s_k})$, $Q_n = \sum_k f''(B_{s_k})(B_{s_{k+1}} - B_{s_k})^2$ and R_n the cubic remainder. Each term has a limit: $S_n \to \mathcal{I}_T(g)$ in L^2 because the step

integrands converge to g in H_T and $\mathcal{I}_T$ is isometric; $Q_n \to \int_0^T f''(B_s)\,\mathrm{d}s$ by the weighted quadratic variation; and $R_n \to 0$ by the remainder estimate. The elementary inequality $(a+b+c)^2 \le 3a^2+3b^2+3c^2$ sends the total error to zero in L^2, so the two sides agree in L^2, hence almost surely. □

Path continuity is supplied by the package's continuous modification and is genuinely used: the Riemann limits run along refining partitions evaluated on paths.

7.6 The time-dependent formula, and the formula as a process

Theorem 7.2 (`ito_formula_td_L2_bddDeriv`). *Let $f\colon \mathbb{R}\times\mathbb{R}\to\mathbb{R}$ have everywhere partial derivatives $f_t, f_x, f_{xx}, f_{tt}, f_{tx}, f_{xxx}$, all bounded, with f_x and f_{xx} jointly continuous. Then there is $g \in H_T$ equal to $f_x(s, B_s(\omega))$ almost everywhere such that, μ-almost surely,*

$$f(T,B_T) - f(0,B_0) = \mathcal{I}_T(g) + \int_{(0,T]} \left(f_t + \tfrac{1}{2}f_{xx}\right)(s,B_s)\,\mathrm{d}s.$$

Two formalization points. First, joint continuity of f_t is not assumed: it is *derived* from the bounded mixed partials (`continuous_uncurry_of_bdd_partials`), keeping the hypothesis package at what a user can check. Second, the time-direction Taylor terms are handled by a boundary-cancellation decomposition in which three telescoping error sums vanish separately, glued by the three-term Cauchy–Schwarz estimate.

The formula also holds as a process. Under the same hypotheses there is an integrand F on the half-line such that, for every $t \le T$, $f(t,B_t) - f(0,B_0) = (F \bullet B)_t + \int_0^t (f_t + \frac{1}{2}f_{xx})(s,B_s)\,\mathrm{d}s$ almost surely, the integral process being a martingale with a continuous local-martingale modification (`ito_formula_td_process`). This statement, unlike Theorem 7.2, does not name F.

7.7 Exponential growth

Bounded derivatives exclude the textbook's first examples: x^2 has an unbounded first derivative, and the geometric Brownian motion value function $S_0\exp((r-\sigma^2/2)t+\sigma x)$ has all its derivatives proportional to $e^{\sigma x}$. The formula is lifted to them by a cutoff localization that reuses the bounded engine.

Theorem 7.3 (`ito_formula_td_localized`). *The conclusion of Theorem 7.2 holds when the six partial derivatives, instead of being bounded, satisfy $|f_\bullet(t,x)| \le Ce^{\lambda|x|}$ for some C and $\lambda \ge 0$, and f_t, f_x, f_{xx} are jointly continuous.*

A smooth truncation ϕ_n, equal to the identity on $[-n,n]$ with bounded derivatives, makes $f_n(t,x) = f(t,\phi_n(x))$ satisfy the bounded hypotheses, so Theorem 7.2 applies to each f_n, with integrands g_n. The boundary and drift terms converge in $L^2(\mu)$ by dominated convergence, the exponential dominators being integrable because Brownian marginals have all exponential moments. Hence the integrals $\mathcal{I}_T(g_n)$ converge and are Cauchy; the isometry transfers Cauchy-ness back to the g_n, which converge in the complete space H_T to some g; continuity of $\mathcal{I}_T$ identifies the limit, and the almost-everywhere identification of g with $f_x(s,B_s)$ passes to the limit along a subsequence. The truncation is the antiderivative of a smooth bump, so every derivative bound comes from Mathlib rather than from explicit calculus. Applied to the exponential, the theorem gives the Itô decomposition of geometric Brownian motion (`ito_formula_gbm`) and, at the risk-neutral drift, the discounted price $S_0e^{-\sigma^2t/2+\sigma B_t}$ as a pure Itô integral with its integrand named: its increment over $[0,T]$ is $\mathcal{I}_T(g)$ with g the class of $\sigma S_0 e^{-\sigma^2 s/2+\sigma B_s}$ (`discountedGBM_eq_itoIntegral`), which is where the library's Black–Scholes layer meets the Itô integral.

7.8 Supporting results

Quadratic variation. For a process with measurable coordinates and centred Gaussian increments of variance $t - s$ (neither independence of increments nor path continuity is assumed), the expected squared-increment sums along equipartitions of $[0, t]$ converge to t (`qv_equals_t`); the L^2 form along uniform partitions is `tendsto_qv`, and convergence in probability is `tendstoInMeasure_qv`. This is convergence of the means; the hypothesis is the Gaussian law of each increment.

Expectation-level Itô. For f with f, f' and f'' bounded and f'' continuous, $\mathbb{E}[f(B_t)] = f(0) + \frac{1}{2}\int_0^t \mathbb{E}[f''(B_s)]\,ds$ for $t > 0$ (`expectation_ito`). It is the bridge between the stochastic tower of this paper and the heat-kernel tower of the parent library, where the Black–Scholes equation is derived through Feynman–Kac.

7.9 Notes on the proof engineering

The density argument is the only place a monotone-class induction was unavoidable in the construction of the integral; expressing it against a trimmed measure, so that the inductively built sets live in the right σ-algebra by construction, is what kept it tractable. The dominated-convergence step that produces the integrand of Itô's formula is delicate because the dominating function must itself be predictable; the trimmed-measure design is what makes the domination land inside H_T without a separate measurability obligation.

8 From L^2 to pathwise: the continuous local martingale

Sections 6 and 7 are an L^2 theory: the process $t \mapsto (\varphi \bullet B)_t$ is continuous *as a curve in* $L^2(\mu)$, and its laws are identities of L^2 elements. The classical object is sharper: for almost every ω it has a continuous path, and it is a continuous local martingale in the pathwise sense. This section builds that object, first on $[0, T]$ and then on the half-line.

8.1 The pathwise gap and the maximal inequality

L^2-continuity in t does not by itself give a continuous path: it controls each time separately, not the trajectory. The bridge is a maximal inequality. `BrownianMotion` supplies Doob's weak-type maximal inequality for right-continuous martingales; applied to the simple integral processes, with the L^1 norm bounded by the L^2 norm and the isometry, it gives $\mu\{\sup_{t \le T} |(V \bullet B)_t| \ge \varepsilon\} \le \varepsilon^{-1}\|V\|_{H_T}$ (`itoSimpleProcess_maximal_prob`). A geometric subsequence makes these tails summable, and Borel–Cantelli makes the approximating step-process integrals converge uniformly on $[0, T]$ for almost every ω. A uniform limit of continuous functions is continuous, so the limit path is continuous off a null set.

8.2 The continuous modification

Theorem 8.1 (`exists_continuous_modification_itoProcess`). *Let B have every path continuous. For every $\varphi \in H_T$ there is a process $\widetilde{X}\colon \mathbb{R}_{\ge 0} \times \Omega \to \mathbb{R}$ whose paths are μ-almost surely continuous on $[0, T]$ such that $\widetilde{X}_t = (\varphi \bullet B)_t$ in $L^2(\mu)$ for every $t \le T$.*

The representative `itoContinuousMod` is the pointwise limit of the approximating step integrals on the good set where they converge uniformly. It equals $(\varphi \bullet B)_t$ almost surely at each t by uniqueness of limits in measure, and its paths are continuous by the uniform-convergence theorem; the supporting bound on the running supremum over $[0, t]$ is `norm_le_iSup_Iic`.

8.3 The local martingale on the null-augmented filtration

A modification fixes the paths but can break adaptedness, since the continuous representative is defined through a limit taken off a null set. The remedy is to complete the filtration: adjoining the μ-null sets to each $\mathcal{F}_t$ gives $\mathcal{F}_t^\mu = \mathcal{F}_t \vee \mathcal{N}$ (`augFiltration`). Completeness is all that is added; right-continuity, the other half of the usual conditions, is neither added nor needed.

Theorem 8.2 (`exists_continuous_localMartingale_modification`). *Let B have every path continuous. For every $\varphi \in H_T$ there is a process X, adapted to $(\mathcal{F}_t^\mu)$ and with everywhere-continuous paths, such that $X_t = (\varphi \bullet B)_t$ in $L^2(\mu)$ for each $t \le T$, and X is a local martingale for $(\mathcal{F}_t^\mu)$.*

Two points fix the content. The underlying L^2 process is a *true* martingale (Section 6); what the theorem adds is an everywhere-continuous representative, in the form the local-martingale interface of `BrownianMotion` takes, whose requirement of paths continuous for *every* ω is what forces an everywhere-defined representative. And the martingale property survives the completion, because a conditional expectation is unchanged when the filtration is enlarged by null sets (`condExp_sup_nulls`).

8.4 The unbounded horizon

The bounded-horizon representatives are consistent: for $m \le n$ the horizon-m and horizon-n continuous processes agree, almost surely and pathwise, on $[0, m)$, because both are continuous and both modify the same L^2 process there. Gluing them gives a single process on the half-line, reading at time t the horizon $\lceil t \rceil + 1$.

Theorem 8.3 (`exists_continuous_localMartingale_modification_infinite`). *For every φ in the unbounded-horizon integrand space there is a process X with everywhere-continuous paths such that $X_t = (\varphi \bullet B)_t$ in $L^2(\mu)$ for every t, and X is a local martingale for $(\mathcal{F}_t^\mu)$ on $\mathbb{R}_{\ge 0}$.*

This is the Itô integral in the form the classical theory gives it: a continuous local martingale on the half-line. The glued process is in fact a true martingale with continuous paths (`exists_continuous_martingale_modification_infinite`), which is what the localization of Section 9 uses.

9 Itô's formula without growth conditions

The L^2 formulas of Section 7 need the integrand $f_x(\cdot, B)$ to lie in H_T, which is what the derivative bounds and growth bounds secure. Without them the stochastic term is no longer an L^2 integral, and the classical statement is that the compensated process is a continuous local martingale. That is the statement proved here, for the classical class $C^{1,2}$.

Theorem 9.1 (`ito_formula_unrestricted`). *Let B have measurable coordinates and continuous paths, and let $f : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ have everywhere partial derivatives f_t, f_x and $f_{xx} = \partial_x f_x$ such that f, f_t, f_x, f_{xx} are jointly continuous. No growth or boundedness condition is imposed. Then there is a process M with every path continuous, which is a local martingale for the null-augmented filtration $(\mathcal{F}_t^\mu)$, such that for every $t \ge 0$, almost surely,*

$$f(t, B_t) - f(0, B_0) = M_t + \int_{(0,t]} \left(f_t + \tfrac{1}{2} f_{xx}\right)(s, B_s)\, \mathrm{d}s.$$

Listing 7. Itô's formula without growth conditions (`Foundations/ItoFormulaUnrestrictedLocMart`).

```
theorem ito_formula_unrestricted (hB : IsPreBrownianReal B μ)
    (hBmeas : ∀ t, Measurable (B t)) (hBcont : ∀ ω, Continuous fun s : ℝ≥0 ↦ B s ω
    ↪ )
    {f f_t f_x f_xx : ℝ → ℝ → ℝ}
    (hf_t : ∀ t x, HasDerivAt (fun s ↦ f s x) (f_t t x) t)
    (hf_x : ∀ t x, HasDerivAt (fun u ↦ f t u) (f_x t x) x)
    (hf_xx : ∀ t x, HasDerivAt (fun u ↦ f_x t u) (f_xx t x) x)
    (hf_cont : Continuous fun p : ℝ × ℝ ↦ f p.1 p.2)
    (hf_t_cont : Continuous fun p : ℝ × ℝ ↦ f_t p.1 p.2)
    (hf_x_cont : Continuous fun p : ℝ × ℝ ↦ f_x p.1 p.2)
    (hf_xx_cont : Continuous fun p : ℝ × ℝ ↦ f_xx p.1 p.2) :
    ∃ M : ℝ≥0 → Ω → ℝ,
      (∀ ω, Continuous fun t ↦ M t ω) ∧
      IsLocalMartingale M (augFiltration (μ := μ) hBmeas) μ ∧
      (∀ t : ℝ≥0, (fun ω ↦ f (t : ℝ) (B t ω) - f 0 (B 0 ω)) =ᵐ[μ]
        (fun ω ↦ M t ω + ∫ s in Set.Ioc 0 t,
          (f_t (s : ℝ) (B s ω) + (1 / 2) * f_xx (s : ℝ) (B s ω)) ∂timeMeasure))
```

The theorem is existential, so as stated it says that the compensated process agrees at each fixed t, almost surely, with a continuous local martingale. The proof takes the compensated process itself as the witness; the statement, not the proof, is what a reader may rely on.

9.1 Exit times as localizers

The localizing sequence is built from the exit time of the closed exterior $\{|x| \geq N\}$, $\tau_N = \inf\{t : |B_t| \geq N\}$ (`exitTime`). Because the hit set is closed and the paths continuous, $\{\tau_N \leq i\}$ is the event $\{\exists s \in [0,i] : |B_s| \geq N\}$, which is measurable for $\mathcal{F}_i$ through a countable rational description. So τ_N is a stopping time for the raw natural filtration, with no right-continuity required; the open exterior $\{|x| > N\}$ would characterise only $\{\tau_N < i\}$, an event of $\mathcal{F}_{i+}$. The package's own localizing sequence of hitting times needs a complete and right-continuous filtration, which the natural filtration is not; the exit time does not. For continuous paths the exit times escape to infinity, since each path is bounded on compacts. The localizer used is $\sigma_N = \min(\tau_N, N)$ (a localizing sequence for the null-augmented filtration, `isLocalizingSequence_sigma`): capping at N makes the time variable range over a compact on $\{t \leq \sigma_N\}$ as well as the space variable.

9.2 Reduction to an Itô process

The witness is $M_t = f(t, B_t) - f(0, B_0) - \int_0^t (f_t + \frac{1}{2} f_{xx})(s, B_s)\, \mathrm{d}s$. It has continuous paths, because f is jointly continuous and the drift primitive is continuous in its upper limit, and the identity of the theorem holds for it by construction. It is adapted to the null-augmented filtration: the drift primitive is $\mathcal{F}_t$-measurable because, after clamping the integrand's time to $[0,t]$, the integrand is a Carathéodory function (continuous in s, $\mathcal{F}_t$-measurable in ω), hence jointly measurable, and its integral is measurable (`StronglyMeasurable.integral_prod_right`). The local-martingale property is not proved by a second truncation of f. It comes from the process form of Itô's formula for an Itô process with unbounded continuous adapted coefficients (Section 10), applied to B itself as the process with $\sigma = 1$ and $b = 0$: that formula supplies a continuous local martingale indistinguishable from M on every $[0,T]$. Stopped at σ_N, the path of B stays in $[-N,N]$ and $t \leq N$, and f, f_t and f_{xx} are bounded on that compact square, so the stopped process is a bounded local

martingale, hence a martingale. These stopped martingales, with the localizing sequence σ_N, assemble into the package's `IsLocalMartingale`.

9.3 What the theorem says, and what it does not

Theorem 9.1 removes the growth conditions of Section 7 entirely, but on its face it says less than the classical formula: M is not named as a stochastic integral $\int_0^t f_x(s, B_s)\,\mathrm{d}B_s$, because the L^2 integral of Section 5 is defined for square-integrable integrands and $f_x(\cdot, B)$ need not be one. Where it is, the identification is Theorem 7.3. For the integrand without any integrability, the stochastic term is the integral of Section 10, built by gluing clipped integrals, and the process formula there names it. The theorem covers every $C^{1,2}$ function, including those that grow faster than any exponential: applied to $f(t,x) = e^{x^2}$, for instance, it gives a continuous local martingale with no integrability of $f(t, B_t)$ assumed.

10 Itô processes

Every model in continuous-time finance is an Itô process $X = x_0 + \int b\,\mathrm{d}s + \int \sigma\,\mathrm{d}B$, so the formula must reach functions of X, not only of (t, B_t). Three results lead there; then comes the formula itself, first for bounded coefficients and then for continuous adapted coefficients without any bound.

Quadratic variation of an Itô process. For $X = X_0 + A + \sigma \bullet B$ with a drift path A Lipschitz in time uniformly in ω, the squared-increment sums along the uniform partition converge to $\int_0^T \sigma_s^2\,\mathrm{d}s$: in L^2 for constant σ (`ItoProcessQV`), and in L^1 for σ bounded, adapted and continuous in time, with the weighted form $\sum_k w(t_k)(\Delta X_k)^2 \to \int_0^T w\sigma^2\,\mathrm{d}s$ for bounded adapted weights with almost surely continuous paths. The drift contributes nothing. The proof splits $(\Delta X_k)^2$ around the frozen increment $\sigma(t_k)\Delta B_k$: the freezing defects $D_k = \Delta M_k - \sigma(t_k)\Delta B_k$ of $M = \sigma \bullet B$ are orthogonal Itô integrals whose sum is the error of the frozen Riemann sum, so $\sum_k \|D_k\|^2 \to 0$ (`AdaptedStochasticIntegralFreezing`), and what remains is the weighted quadratic variation of B. The L^1 mode is a limitation of the method: an L^2 bound on the freezing part would need fourth moments of $\sigma \bullet B$.

Riemann–Stieltjes sums against an Itô integral. For $M = \varphi \bullet B$ and a bounded predictable weight w whose paths are almost surely continuous on $[0, T]$, the sums $\sum_k w(t_k)(M_{t_{k+1}} - M_{t_k})$ converge in mean square to the integral of w against M (Section 11.2), which is the Itô integral against B of the class of φw (`AdaptedRiemannStieltjes`). The route goes through $L^2(\langle M\rangle)$: each sum is the integral against M of a left-endpoint step process, the step processes converge to w by dominated convergence, and the isometry against M carries the convergence over. No path regularity of φ is needed.

Constant coefficients. For $X_t = X_0 + bt + \sigma B_t$ and f whose first three derivatives grow at most exponentially, the formula $f(X_T) - f(X_0) = \mathcal{I}_T(\sigma f'(X)) + \int_0^T (f'(X_s)b + \frac{1}{2}f''(X_s)\sigma^2)\,\mathrm{d}s$ holds with the integrand named (`ito_formula_itoProcess`); this is the form the geometric Brownian motion needs.

Theorem 10.1 (`ito_formula_adapted`). *Let B have every path continuous, let σ and b be bounded, adapted to the natural filtration of B, with every path continuous, and let $X = x_0 + \int_0^{\cdot} b\,\mathrm{d}s + \int_0^{\cdot} \sigma\,\mathrm{d}B$. For f twice differentiable with f', f'' bounded and f'' continuous, almost surely*

$$f(X_T) - f(x_0) = \mathcal{I}_T\big(f'(X)\,\sigma\big) + \int_0^T \Big(f'(X_s)\,b_s + \tfrac{1}{2}f''(X_s)\,\sigma_s^2\Big)\mathrm{d}s,$$

the stochastic integral being the Itô integral at horizon T of the class of the bounded predictable process $f'(X)\sigma$.

The process X is defined (`adaptedItoProcess`) as x_0 plus the pathwise drift integral plus the continuous modification of the Itô integral process of σ. The proof applies the discrete Itô formula on the uniform partition, which splits $f(X_T) - f(x_0)$ into drift sums, Riemann–Stieltjes sums against $M = \sigma \bullet B$, weighted quadratic variation and a remainder. The three random limits hold in measure; a common subsequence along which all three converge almost surely is extracted, and on such a path, where X is also continuous, a deterministic lemma (`ito_formula_td_of_tendsto`) passes the discrete identity to the limit. The limit of the Riemann–Stieltjes sums is the integral against M of $f'(X)$, which is the L^2 integral against B of the class of $f'(X)\sigma$ (Section 11.2), the displayed form. Because the limit is taken almost surely along a subsequence rather than in L^2 at a rate, the Taylor remainder is controlled by the modulus of continuity of f'' times the quadratic variation, and no third derivative is needed.

10.1 Without bounds: the integral by gluing, and the formula as an identity of processes

A continuous adapted σ need not be bounded, nor $\int_0^T \sigma^2 \,\mathrm{d}s$ integrable, yet every path is bounded on $[0, T]$. The integral is built by localizing in the sample space, without stopping times. Clip σ to $[-N, N]$; the clipped integrand is bounded, adapted and path-continuous, so it has the continuous Itô integral M^N. On the event that the path of σ stays in $[-n, n]$ up to t, the integrands $\mathrm{clip}_n\sigma$ and $\mathrm{clip}_N\sigma$ ($n \le N$) agree on $[0, t]$, and so do their left-endpoint Riemann sums, which converge in measure to M_t^n and M_t^N; limits in measure are local, so $M_t^n = M_t^N$ almost surely on that event, and by continuity at every such t at once. The glued integral (`localItoIntegral`) reads, at time t and on each path, M_t^N at the least integer N bounding the path on $[0, t]$. It is adapted to the natural filtration, almost surely continuous on $[0, T]$, equal to the L^2 integral when σ is bounded, independent of the horizon up to indistinguishability (`ae_localItoIntegral_eq_of_le`), and the limit in probability of the left-endpoint Riemann sums. It has a version with every path continuous that is a local martingale for the null-augmented filtration, localised by the exit times of σ (`exists_continuous_localMartingale_localItoIntegral`). The same construction for a predictable integrand whose paths are continuous only almost surely, such as the integrand $f_x(s, X_s)\sigma_s$ of Itô's formula, is `predItoIntegral`; it reads the level on a countable dense set of times, agrees with `localItoIntegral` up to indistinguishability on $[0, T]$ where both are defined (`ae_predItoIntegral_eq_localItoIntegral`), and has a continuous local-martingale version (`exists_continuous_localMartingale_predItoIntegral`).

Theorem 10.2 (`ito_formula_adapted_unbounded_process`). *Let B have every path continuous, let σ and b be adapted to the natural filtration of B with every path continuous, not necessarily bounded, and let $X = x_0 + \int_0^{\cdot} b \,\mathrm{d}s + \int_0^{\cdot} \sigma \,\mathrm{d}B$ (`localItoProcess`), the stochastic integral being `localItoIntegral`. Let $f(t, x)$ have everywhere partial derivatives f_t, f_x, f_{xx}, jointly continuous, with no bound on f or on them. Then, almost surely, for all $t \in [0, T]$ at once,*

$$f(t, X_t) = f(0, x_0) + \int_0^t \left(f_t + f_x b + \tfrac{1}{2} f_{xx}\sigma^2\right)(s, X_s)\,\mathrm{d}s + \left(\int f_x(\cdot, X)\,\sigma \,\mathrm{d}B\right)_t,$$

the last term being `predItoIntegral` on $[0, T]$ of the predictable integrand $f_x(s, X_s)\sigma_s$.

For $f(x)$ twice differentiable with f'' continuous this is `ito_formula_adapted_unbounded_C2_process`, the case $f_t = 0$. The stochastic term is a continuous local martingale, so $f(t, X_t)$ minus its drift integral has a version with every path continuous that is a local martingale for the null-augmented filtration, equal to it almost surely at each

$t \le T$ (`exists_localMartingale_ito_formula`). The statement is about the horizon T fixed in advance, a starting point x_0 that is deterministic, and a process X built from B and not an arbitrary semimartingale.

The proof has three steps. At each $t \le T$, the Riemann–Stieltjes sums of $f_x(\cdot, X)$ against $M = \int \sigma \, \mathrm{d}B$ converge in probability to $f(t, X_t) - f(0, x_0) - \int_0^t (f_t + f_x b + \frac{1}{2} f_{xx} \sigma^2)(s, X_s) \, \mathrm{d}s$. For bounded coefficients this is the bounded-case argument after a cutoff of f to a compact square; for unbounded ones, on the event that σ and b stay in $[-N, N]$ up to T the process is the one built from the clipped coefficients, with the same Riemann sums and the same limit, and these events exhaust Ω in probability (`ito_formula_adapted_unbounded_le`). The same sums converge in probability to $\left(\int f_x(\cdot, X)\sigma \, \mathrm{d}B\right)_t$ by the construction above, so the two sides agree almost surely at each t. Both have almost surely continuous paths on $[0, T]$, hence agree at all t at once.

Theorem 10.3 (`ito_product_rule_process`). *Let B have every path continuous and let σ, τ, b, c be adapted to its natural filtration with every path continuous. For $X = x_0 + \int b \, \mathrm{d}s + \int \sigma \, \mathrm{d}B$ and $Y = y_0 + \int c \, \mathrm{d}s + \int \tau \, \mathrm{d}B$ as above, almost surely, for all $t \in [0, T]$ at once,*

$$X_t Y_t = x_0 y_0 + \int_0^t \left(X_s c_s + Y_s b_s + \sigma_s \tau_s\right) \mathrm{d}s + \left(\int (X\tau + Y\sigma) \, \mathrm{d}B\right)_t,$$

the stochastic integral being `predItoIntegral`.

In differential form $\mathrm{d}(XY) = X \, \mathrm{d}Y + Y \, \mathrm{d}X + \sigma\tau \, \mathrm{d}t$, and $\int_0^t \sigma\tau \, \mathrm{d}s$ is the covariation of X and Y, derived rather than assumed. The proof is polarisation: the formula for $f(x) = x^2/2$ along X, Y and $Z = X + Y$ gives $XY = \frac{1}{2}(Z^2 - X^2 - Y^2)$, where Z is again an Itô process, with coefficients $\sigma + \tau$ and $b + c$, because both integrals are additive up to indistinguishability, being limits in probability of Riemann sums that add. The product rule is stated only for $f(x, y) = xy$ and one scalar driver; a general $f(t, X, Y)$ with cross term $f_{xy}\sigma\tau$ needs the Taylor argument in two coordinates and is not formalized.

11 The integral as an operator: its range and its uniqueness

Two theorems characterise $\mathcal{I}_T$ as an operator. Its range is as large as it can be, and it is the only reasonable integral.

11.1 Martingale representation

Theorem 11.1 (`exists_itoIntegral_representation`). *Let B have every path continuous. For every $F \in L^2(\mu)$ that is $\mathcal{F}_T^B$-measurable there is a unique $\varphi \in H_T$ with*

$$F = \mathbb{E}[F] + \mathcal{I}_T(\varphi) \qquad \mu\text{-a.s.}$$

Equivalently, the range of $\mathcal{I}_T$ together with the constants is the whole of $L^2(\mathcal{F}_T^B)$ (`itoIntegralCLM_surjective_onto_centered`), and $\mathcal{I}_T$ is an isometric isomorphism of H_T onto the centred part of $L^2(\mathcal{F}_T^B)$ (`itoIsometryEquiv`).

Listing 8. Martingale representation, terminal form (`Foundations/MartingaleRepresentation`).

```
theorem exists_itoIntegral_representation (hB : IsPreBrownianReal B μ)
    (hBmeas : ∀ t, Measurable (B t))
    (hBcont : ∀ ω, Continuous fun s : ℝ≥0 ↦ B s ω) (T : ℝ≥0)
    (F : Lp ℝ 2 μ)
    (hFmeas : AEStronglyMeasurable[ItoIntegralL2.natFiltration hBmeas T] (⇑F) μ) :
    ∃! φ : Lp ℝ 2 (trimMeasure (μ := μ) T hBmeas),
      (⇑F) =ᵐ[μ] fun ω ↦ (∫ x, F x ∂μ) + itoIntegralCLM hB T hBmeas φ ω
```

The process form follows from Theorem 6.1: every square-integrable martingale for the natural filtration is $M_t = M_0 + (\varphi \bullet B)_t$ for $t \le T$ (`martingale_representation`), because $(\varphi \bullet B)_t$ is the $\mathcal{F}_t$-conditional expectation of $\mathcal{I}_T(\varphi)$.

The proof uses no Malliavin calculus, no Clark–Ocone formula and no Itô formula for an adapted integrand. The range of $\mathcal{I}_T$ is closed, being the image of an isometry, so it suffices to show that a centred $\xi \in L^2(\mathcal{F}_T^B)$ orthogonal to the range is zero. Two facts do this. First, for every step function h the Doléans exponential $\mathcal{E}(h) = \exp(\int h \,\mathrm{d}B - \frac{1}{2}\int h^2 \,\mathrm{d}s)$ satisfies $\mathcal{E}(h) - 1 \in \operatorname{range} \mathcal{I}_T$ (`stepDoleans_sub_one_mem_range`), starting from the constant step, which is the discounted geometric Brownian motion of Section 7.7, and adding one step at a time with the locality of the integral. Second, an element orthogonal to every $\mathcal{E}(h)$ vanishes (`eq_zero_of_orthogonal_stepDoleans`). Abel summation turns the step exponentials into the exponentials $\exp(\sum_i \lambda_i B_{t_i})$ of finitely many coordinates; on each dyadic cylinder σ-algebra the conditional expectation of an orthogonal element then has a vanishing Laplace transform, which forces it to zero by analytic continuation of the complex moment-generating function and uniqueness of characteristic functions; and by path continuity the dyadic cylinder σ-algebras increase to $\mathcal{F}_T^B$ (`iSup_cylinderFiltration_eq_natFiltration`), so Lévy's upward theorem sends the conditional expectations to ξ itself. Centring, $\mathbb{E}[\mathcal{I}_T(\varphi)] = 0$, is proved a floor down (`integral_itoIntegralCLM`).

In the parent library this theorem is market completeness: every square-integrable $\mathcal{F}_T^B$-measurable claim is attained from initial wealth $\mathbb{E}[H]$ by a unique strategy (`exists_replicating_strategy`). What is not proved is the explicit form of φ, which is the Clark–Ocone formula and needs the Malliavin derivative.

11.2 Integration against an Itô integral

For $\varphi \in H_T$ let $M = \varphi \bullet B$. The bracket of M is the measure $\langle M \rangle = \varphi^2 \cdot (\lambda_T \otimes \mu)$ on the predictable σ-algebra (`bracketMeasure`), and the integral against M is defined on $L^2(\langle M \rangle)$ by composing $\mathcal{I}_T$ with the multiplication isometry $\psi \mapsto \psi\varphi$ from $L^2(\langle M \rangle)$ to H_T:

$$\int \psi \,\mathrm{d}M = \mathcal{I}_T(\psi\varphi), \qquad \Big\| \int \psi \,\mathrm{d}M \Big\|_{L^2(\mu)} = \|\psi\|_{L^2(\langle M \rangle)}$$

(`itoIntegralAgainstCLM`, `norm_itoIntegralAgainstCLM`). The chain rule is thus how the object is built; Mathlib has the multiplication isometry only for $p = 1$, so the $p = 2$ case is proved here (`LpMulIsometry`). What identifies the result as *the* integral against M is the Riemann–Stieltjes value on elementary integrands, $\int Z\mathbf{1}_{(a,b]} \,\mathrm{d}M = Z(M_b - M_a)$ for bounded $\mathcal{F}_a$-measurable Z (`itoIntegralAgainst_elementary`), together with uniqueness among continuous linear maps with those values (`itoIntegralAgainst_unique`).

The bracket earns its name. The second moment of an increment is its mass, $\mathbb{E}[(M_b - M_a)^2] = \langle M \rangle((a,b] \times \Omega)$ (`norm_sq_increment_eq_bracket`); conditionally, $\mathbb{E}[(M_b - M_a)^2 \mid \mathcal{F}_a] = \mathbb{E}[\int_a^b \varphi_u^2 \,\mathrm{d}u \mid \mathcal{F}_a]$ against the pathwise increment of the bracket (`condExp_band_second_moment`); and that increment is adapted (`bracketProcess_adapted`), so $\mathbb{E}[M_b^2 - \langle M \rangle_b \mid \mathcal{F}_a] = M_a^2 - \langle M \rangle_a$ (`condExp_sq_sub_bracket`): the bracket is the compensator of M^2. Not formalized are any pathwise quadratic variation of M, a bundled martingale structure for $M^2 - \langle M \rangle$, and the Doob–Meyer decomposition of a general submartingale.

11.3 Agreement with the upstream characterisation

`BrownianMotion` characterises a stochastic integral axiomatically [2]. A map I from processes to random variables, defined on a domain S, is a *Riemann–Stieltjes extension* if it returns $X(Y_j - Y_i)$ on every elementary process $\mathbf{1}_{(i,j]}X$ with simple $\mathcal{F}_i$-measurable X, is

linear, respects indistinguishability, and passes to the limit under dominated convergence; it is a *stochastic integral* if, in addition, it agrees almost surely with every Riemann–Stieltjes extension on the intersection of their domains.

THEOREM 11.2 (`isStochasticIntegral`). *For every $\varphi \in H_T$, the integral against $M = \varphi \bullet B$, with integrator M stopped at T, the natural filtration, and as domain the processes indistinguishable from a predictable process square-integrable against $\langle M \rangle$, is a stochastic integral in the sense of* `BrownianMotion`.

Listing 9. The characterisation (`Foundations/StochasticIntegralCharacterisation`).

```
theorem isStochasticIntegral (hB : IsPreBrownianReal B μ) :
    IsStochasticIntegral μ (integrator T hBmeas φ hB) (natFiltration hBmeas)
      (sIntegral T hBmeas φ hB) (domain T hBmeas φ)
```

Taking $\varphi = 1$ makes M the Brownian motion stopped at T, so the theorem says in particular that $\mathcal{I}_T$ is a stochastic integral against B in the upstream sense. Two features of the proof are worth recording. The domain is forced rather than chosen: the predicate asks for closure under indistinguishability, and a process indistinguishable from a predictable one need not be predictable, or even jointly measurable, so the domain is the processes with a predictable square-integrable version, and the integral reads the class of any version. And the uniqueness clause is the substantive part. An extension in the upstream sense is not assumed to be L^2-continuous, so uniqueness among continuous linear maps does not reach it; the proof is a monotone-class argument from dominated convergence alone, through a Dynkin argument over the predictable rectangles, then simple functions, bounded processes, and finally the whole common domain by truncation. Dominated convergence in L^p, which Mathlib at the pin has only for $p = 1$, is proved on the way for every $p < \infty$ and every measure (`tendsto_eLpNorm_sub_of_dominated_convergence`).

12 What the calculus carries

The integral and the formula are consumed downstream. The results below belong to stochastic analysis proper; the library's pricing layers use them in the companion papers [4, 3].

Lévy's characterisation. Let $\mathcal{G}$ be a filtration indexed by $\mathbb{R}_{\geq 0}$ and M a process adapted to it, with every path continuous and $M_0 = 0$ everywhere, such that M and $M^2 - t$ are local martingales in the sense of `BrownianMotion` (`isBrownianReal_of_localMartingale`). Then M is a Brownian motion, and moreover every increment $M_t - M_s$ is independent of $\mathcal{G}_s$ (`isFilteredPreBrownian_of_localMartingale`). The bracket enters only as the hypothesis that $M^2 - t$ is a local martingale: no quadratic-variation process is constructed and no stochastic integral against M is used. The proof stops M at the exit times of $|M|$, so that M is a bounded martingale with $M^2 - (t \wedge \rho_n)$ a martingale; a cell-by-cell expansion along uniform partitions makes $\exp(aM - \frac{a^2}{2}(t \wedge \rho_n))$ a martingale for every a, and a limit, using an L^2 bound and Vitali, gives the exponential martingale $\exp(aM_t - a^2 t/2)$, hence Brownian motion by characteristic functions. Processes whose paths are continuous only almost surely, or started at zero only almost surely, and the multi-dimensional statement are not covered.

Girsanov's theorem. For a bounded predictable drift θ, $|\theta| \leq C$, and a horizon T, let $\frac{\mathrm{d}Q}{\mathrm{d}\mu} = \exp\big(-\mathcal{I}_T(\theta) - \frac{1}{2}\int_0^T \theta_s^2 \,\mathrm{d}s\big)$ and $B^\theta_u = B_u + D_u$ with D the continuous modification of the drift integral $\int_0^u \theta_s \,\mathrm{d}s$ (frozen at its value at T after T). Under Q, B^θ

starts at 0, has increments $B^\theta_t - B^\theta_s \sim N(0, t-s)$, and every increment is independent of $\mathcal{F}^B_s$, for all $s \le t$ in $\mathbb{R}_{\ge 0}$, so B^θ is a pre-Brownian motion for the natural filtration of B (`Btheta_isFilteredPreBrownian_predictable_of_bdd`). If moreover B has almost surely continuous paths, then so has B^θ under Q, and B^θ is a Brownian motion (`Btheta_isBrownianReal_predictable_of_bdd`). An earlier version of this statement fixed only the finite-dimensional laws of B^θ on $[0, T]$ (`Btheta_isQBrownianMotion_predictable_of_bdd`); that result is the last rung of a ladder (constant, simple adapted, continuous adapted, bounded predictable θ) whose rungs share one lemma: any process supplying its own exponential martingale under Q has the Brownian finite-dimensional laws on $[0, T]$ (`isQBrownianMotion_of_expMartingale`), by characteristic functions, with no adapted-integrand Itô formula. The measure Q depends on T. The general case under Novikov's condition is not formalized.

Stochastic differential equations. For Lipschitz b and σ, with constants L_b and L_σ, and B with every path continuous, the Picard map $X \mapsto \eta + \int_0^\cdot b(X)\,\mathrm{d}s + \int_0^\cdot \sigma(X)\,\mathrm{d}B$ acts on the predictable L^2 space H_T. Some iterate is a contraction, with factorial decay of the iterates, so the map has exactly one fixed point on every horizon T, with no smallness condition (`picardMap_exists_unique_fixedPoint_any_horizon`). When $TL_b + \sqrt{T}\,L_\sigma < 1$ the map itself is a contraction with the same conclusion (`picardMap_exists_unique_fixedPoint`), and only then is the fixed point realized as a sample-path process: almost everywhere on the space–time product, $X = \eta + \int_0^\cdot b(X)\,\mathrm{d}s + (\sigma(X) \bullet B)$ with the drift identified as a pathwise Lebesgue integral and the diffusion term as the continuous modification of the Itô integral process (`sde_pathwise_decomposition`, `driftContinuousMod_eq_setIntegral`). The solution on a long horizon is therefore an element of L^2, an equivalence class of processes, not a sample-path process. A separate uniqueness lemma, by an energy estimate and Grönwall's inequality, shows that two processes satisfying the equation with square-integrable differences agree almost surely at each time (`sde_pathwise_uniqueness`); in that statement the diffusion term is an abstract operator whose Itô isometry bound is assumed, and the lemma is not connected to the Picard fixed point. For geometric Brownian motion a sample-path solution is exhibited directly: for the Black–Scholes asset path S, almost surely and for all $t \le T$ at once, $S_t = S_0 + \int_0^t rS_s\,\mathrm{d}s + (\int \sigma S\,\mathrm{d}B)_t$, with the stochastic integral `predItoIntegral` (`pathAssets_solves_sde`).

Stochastic Fubini. Let m be a finite measure, $I\colon L^2(m) \to F$ any continuous linear map into a Banach space, $\varphi\colon A \to L^2(m)$ Bochner integrable against an s-finite measure ν, ψ a jointly integrable family of representatives of φ, and $g \in L^2(m)$ a representative of $x \mapsto \int \psi(u,x)\,\nu(\mathrm{d}u)$. Then $\int I(\varphi(u))\,\nu(\mathrm{d}u) = I(g)$ (`stochFubini_ofCLM`). The Itô integral is an instance (`stochFubini_ito`), and the Wiener integral through its identification with $\mathcal{I}_T$; no property of I beyond continuity and linearity is used. This is the Fubini theorem for parameter integrals of stochastic integrals of L^2 integrands, not for integrands with only a.s. finite energy.

An application: the delta hedge. Let B have every path continuous, $S_t = S_0 \exp((r - \sigma^2/2)t + \sigma B_t)$, $K, \sigma, S_0 > 0$, $T' < T$ and $C(t,S)$ the Black–Scholes call price with time $T-t$ to maturity. Almost surely, for all $t \le T'$ at once, $e^{-rt}C(t,S_t) = C(0,S_0) + \int_0^t \Phi(d_1)\,\sigma e^{-rs}S_s\,\mathrm{d}B_s$, the integral being `predItoIntegral` of the delta times the diffusion coefficient of the discounted price (`bs_delta_hedge_replication`). It is Theorem 10.2 applied to the discounted call value in the Brownian coordinate, whose drift vanishes by the Black–Scholes equation; the horizon $T' < T$ is needed because the call price is singular at maturity.

13 Scope: what is, and is not, an Itô calculus here

The title claims "a" calculus, and the indefinite article is doing deliberate work. What earns the noun is that the canonical objects and theorems are present and *compose*: the stochastic term of Itô's formula is the isometric integral of Section 5, whose energy law is the isometry of Section 6, whose conditional projection is the martingale of Theorem 6.1, whose continuous representative is the local martingale of Section 8, whose range is characterised by Theorem 11.1, and which is the upstream package's stochastic integral by Theorem 11.2. One object, a single chain of theorems, no re-derivation gluing them. What the indefinite article concedes is equally concrete.

The integrator. The integrator is Brownian motion, or an Itô integral against it. Integration against a general continuous local martingale or semimartingale is not formalized; Theorem 11.2 places the present integral inside the upstream package's axiomatic characterisation of a stochastic integral.

Integrands. Integrands live in L^2: in H_T on $[0, T]$, or in its half-line analogue. Beyond L^2 the development has the integral of a predictable integrand with almost surely continuous paths, built by gluing clipped integrals (Section 10.1); the localized class $\{\varphi : \int_0^T \varphi_s^2 \, \mathrm{d}s < \infty \text{ a.s.}\}$ in general is not covered.

The scope of each Itô formula. In L^2, the formula for $f(t, B_t)$ needs the six partial derivatives bounded or of exponential growth in x. In local-martingale form (Theorem 9.1) it needs f_t, f_x, f_{xx} continuous and nothing else, and the statement is existential: the compensated process agrees at each t with a continuous local martingale. For Itô processes with adapted coefficients it is proved as an identity of processes for $f(t, x)$ with continuous f_t, f_x, f_{xx} (Theorem 10.2), with continuous adapted σ and b and no bound on them, at a horizon T fixed in advance; the stochastic term is the glued integral, not an L^2 integral. It is not proved for a random initial value, on $T = \infty$, for a semimartingale not built from B, or for a function of two such processes other than the product xy (Theorem 10.3). The multi-dimensional formula is recorded in the library only as a reduced core, whose hypotheses include its conclusion.

Completeness, not the usual conditions. The pathwise local martingales live on the null-augmented filtration: completeness, which the continuous modification needs, is added, but right-continuity, the other half of the usual conditions, is neither assumed nor used.

Quadratic variation in mean, not pathwise. Quadratic variation is established in L^1 and L^2 along uniform or refining partitions, for Brownian motion and for Itô processes, and the bracket of $\varphi \bullet B$ is identified in conditional mean. No pathwise (almost sure) quadratic variation is formalized; Lévy's characterisation avoids it by taking $M^2 - t$ to be a local martingale as a hypothesis.

Representation, Girsanov, equations. The explicit form of the representing integrand (Clark–Ocone) is open. Girsanov's theorem is proved for bounded predictable drift; the Novikov-condition version and the general Girsanov theorem are recorded only as reduced cores. Stochastic differential equations have existence and uniqueness on every horizon in the predictable L^2 space, but a sample-path solution is constructed only on short horizons, where the Picard map is a contraction; and the converse direction of the continuous-time fundamental theorem of asset pricing is open in the parent library [3].

None of these boundaries qualifies a statement made in Sections 5 to 12; they bound the *reach* of the development, not the *validity* of any theorem in it.

14 Related work

Keskin's Isabelle/HOL martingale library [10] is the closest prior art in the probabilistic direction; it develops sub- and supermartingales in discrete and continuous time and stops below stochastic integration. Degenne, Ledvinka, Marion and Pfaffelhuber [5, 2] construct Brownian motion in Lean, through the Kolmogorov extension theorem, Gaussian measures and a Kolmogorov–Chentsov theorem; their package is the foundation of this work, and its stochastic-integration programme (an axiomatic characterisation of the integral, local martingales, quadratic variation of locally square-integrable martingales, and the Doob–Meyer decomposition) is complementary to it: a general theory under construction, against which Theorem 11.2 certifies the concrete Brownian integral built here. Nagy's Lean derivation from Itô's lemma to Black–Scholes [12] formalizes the structural drift computation of Itô's lemma and, on its own account, leaves the continuous Itô integral structurally verified rather than constructed. Avigad, Hölzl and Serafin's central limit theorem [1] remains the historical anchor for deep measure-theoretic probability in proof assistants, and Echenim, Guiol and Peltier formalized discrete-time replication pricing in Isabelle/HOL [7]. StochBench [13] is a recent Lean benchmark of graduate stochastic-process problems which, lacking infrastructure such as the Itô integral, states many of its targets with the required properties as hypotheses. A search in October 2026 of the Isabelle Archive of Formal Proofs and the Lean and Coq/Rocq ecosystems found no other completed machine-checked construction of the Itô integral, proof of Itô's formula, or martingale representation theorem.

15 Conclusion

The development formalizes the Itô calculus of Brownian motion: the integral on the half-line as an isometry, with the integral on $[0, T]$ as its restriction; the integral as a continuous L^2 martingale with its full second-moment structure, its range (martingale representation) and its uniqueness (agreement with the upstream axiomatic integral); Itô's formula in L^2 under bounded or exponentially growing derivatives, as a process, and for Itô processes with continuous adapted coefficients, bounded or not, as an identity of processes; and the pathwise object, a continuous local martingale on the half-line, to which Itô's formula extends without any growth condition. To our knowledge it is the first machine-checked Itô calculus in any proof assistant. It is also the layer the parent library [4] stands on, where the integral and the formula are consumed to construct, rather than assume, objects of continuous-time pricing: the dynamic change of measure, the replicating hedge, and the discounted price as an Itô integral. The boundaries of Section 13 (general semimartingale integrators, the localized integrand class, pathwise quadratic variation, and the multi-dimensional formula) mark where the formalized theory currently ends.

Provenance. The Lean development was written by the author with an AI coding assistant (Claude Code), as recorded in the repository's provenance file; every proof is checked by the Lean kernel and passes the gates of Section 2 unchanged. This manuscript was revised with the same assistant, and its statements were checked against the Lean sources of release `v1.11.0`.

A Displayed results and their Lean names

Result	Lean name	Module / benchmark entry
Rectangle π-system	`isPiSystem_predictableRect`	`ItoIntegralHorizonDense`
Generation	`generateFrom_predictableRect`	`ItoIntegralHorizonDense`
Density	`simpleAssemblyHorizon_denseRange`	`ItoIntegralHorizonDense`
Unbounded horizon	`itoIntegralL2`, `itoIntegralL2_norm`	`ItoIntegralL2Dense` / `sc-ito-infinite-horizon-isometry`
The Itô integral, isometry	`itoIntegralCLM`, `itoIntegralCLM_norm`	`ItoIntegralCLM`
Keystone $\mathcal{I}_T(g_B)$	`itoIntegralCLM_brownian`	`ItoIntegralBrownian`
Wiener integral is the Itô integral	`wienerIntegralLp_eq_itoIntegralCLM`	`WienerIntegralIto` / `mf-wiener-integral-is-ito-integral`
Conditional increments	`condExp_adapted_mul_increment`	`ItoIntegralProcessMartingale`
Key identity (Theorem 6.1)	`itoProcessCLM_eq_condExpL2`	`ItoIntegralProcessGeneral`
Martingale property	`itoIntegralProcessGen_isMartingale`	`ItoIntegralProcessGeneral`
Time-indexed isometry	`itoProcessCLM_norm_sq`	`ItoIntegralProcessIsometry` / `sc-ito-general-time-isometry`
QV in L^2, in mean	`tendsto_qv`, `qv_equals_t`	`QuadraticVariationL2`, `BrownianQuadraticVariation` / `sc-thm-6.1.1`
Weighted QV	`tendsto_weighted_qv_process`	`WeightedQuadraticVariation`
Itô, bounded (Theorem 7.1)	`ito_formula_L2_bddDeriv`	`ItoFormulaCLM` / `sc-thm-7.1.1`
Itô, time-dependent (Theorem 7.2)	`ito_formula_td_L2_bddDeriv`	`ItoFormulaTD` / `sc-thm-7.1.2`
Itô as a process	`ito_formula_td_process`	`ItoFormulaProcess` / `sc-ito-formula-td-process`
Itô, exponential growth (Theorem 7.3)	`ito_formula_td_localized`	`ItoFormulaLocalized` / `sc-ito-formula-localized`
GBM; discounted GBM	`ito_formula_gbm`, `discountedGBM_eq_itoIntegral`	`ItoFormulaGBM` / `sc-ito-formula-gbm`, `sc-discounted-gbm-ito`
Expectation Itô	`expectation_ito`	`FeynmanKacHeatEquation`
Continuous modification	`exists_continuous_modification_itoProcess`	`ItoIntegralProcessContinuousModification`
Local martingale on $[0, T]$	`exists_continuous_localMartingale_modification`	`ItoIntegralProcessLocalMartingaleGeneral`
Local martingale on $\mathbb{R}_{\geq 0}$	`exists_continuous_localMartingale_modification_infinite`	`ItoIntegralProcessLocalMartingaleInfinite` / `sc-ito-infinite-local-martingale`
Itô, no growth (Theorem 9.1)	`ito_formula_unrestricted`	`ItoFormulaUnrestrictedLocMart` / `sc-ito-formula-unrestricted-islocalmartingale`
Exit times	`exitTime`	`ExitTime`
Itô, constant coefficients	`ito_formula_itoProcess`	`ItoFormulaItoProcess` / `sc-ito-formula-ito-process`
Itô, bounded adapted coefficients (Theorem 10.1)	`ito_formula_adapted`	`ItoFormulaAdaptedTD` / `sc-ito-formula-adapted`

Result	Lean name	Module / benchmark entry
Integral, unbounded continuous adapted	`exists_continuous_localMartingale_localItoIntegral`	`ItoIntegralLocalisedLocalMartingale` / `sc-ito-integral-unbounded-local-martingale`
Integral, predictable a.s. continuous	`exists_continuous_localMartingale_predItoIntegral`	`ItoIntegralLocalisedPredictable` / `sc-ito-integral-predictable-ae-continuous`
Itô process, unbounded (Theorem 10.2)	`ito_formula_adapted_unbounded_process`, `ito_formula_adapted_unbounded_C2_process`, `exists_localMartingale_ito_formula`	`ItoFormulaAdaptedProcess` / `sc-ito-formula-adapted-process`
Product rule (Theorem 10.3)	`ito_product_rule_process`	`ItoProductRule` / `sc-ito-product-rule`
Representation (Theorem 11.1)	`exists_itoIntegral_representation`, `itoIntegralCLM_surjective_onto_centered`, `itoIsometryEquiv`	`MartingaleRepresentation`
Representation, process form	`martingale_representation`	`MartingaleRepresentation` / `gir-thm-9.3.4`
Integral against $\varphi \bullet B$	`itoIntegralAgainstCLM`, `itoIntegralAgainst_elementary`	`ItoIntegralAgainstMartingale` / `sc-ito-integral-band`
Bracket, conditional	`condExp_band_second_moment`	`PointwiseBracket` / `sc-bracket-conditional`
Characterisation (Theorem 11.2)	`isStochasticIntegral`	`StochasticIntegralCharacterisation` / `sc-ito-is-stochastic-integral`
Lévy's characterisation	`isBrownianReal_of_localMartingale`, `isFilteredPreBrownian_of_localMartingale`	`LevyCharacterization` / `sc-thm-9.1.1`
Girsanov, finite-dimensional laws	`Btheta_isQBrownianMotion_predictable_of_bdd`	`GirsanovPredictableTheta` / `gir-thm-9.1.8-predictable`
Girsanov, for the filtration	`Btheta_isFilteredPreBrownian_predictable_of_bdd`, `Btheta_isBrownianReal_predictable_of_bdd`	`GirsanovFiltered` / `gir-filtered-brownian-predictable`, `gir-brownian-continuous-predictable`
SDE, short horizon	`picardMap_exists_unique_fixedPoint`, `sde_pathwise_decomposition`	`SDEExistence`, `SDEPathwise` / `sde-picard-existence-uniqueness`, `mf-sde-picard-pathwise`
SDE, every horizon	`picardMap_exists_unique_fixedPoint_any_horizon`	`SDEExistence` / `sc-sde-existence-any-horizon`
SDE, uniqueness at each time (abstract)	`sde_pathwise_uniqueness`	`SDEUniqueness`
Black–Scholes path solves its SDE	`pathAssets_solves_sde`	`Contracts/Models/BlackScholesSDE` / `mf-bs-path-solves-sde`
Stochastic Fubini	`stochFubini_ofCLM`, `stochFubini_ito`	`StochasticFubini`, `StochasticFubiniIto` / `sc-stochastic-fubini-clm`, `sc-stochastic-fubini-ito`
Delta hedge	`bs_delta_hedge_replication`	`BlackScholes/DeltaHedge` / `mf-bs-delta-hedge-replication`

All modules are under `MathFin/Foundations/` except those given with a directory. The library contains no `sorry`. Every theorem in the table is pinned directly by one of the two axiom audits to the three standard axioms, except the supporting lemma `isPiSystem_predictableRect`, which is used in the proofs of pinned theorems; definitions are not pinned separately.

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